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Modeling

- Linear optimization

$$\min\ c^T x$$
$$s.t\ \ Ax \geq b$$

- Quadratic optimization

$$\min\ c^T x + x^T Q x$$
$$s.t\ \ Ax \geq b$$

e.g., $\min_x \sum_{i=1}^{m} (y_i - a_i^T x)^2$

- Mixed-integer (qp) optimization (quadratic / linear)

$$\min\ c^T x + x^T Q x$$
$$s.t$$

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- Quadratic optimization

$$\min \; c^T x + x^T Q x$$
$$s.t \quad Ax \geq b$$

$$e.g., \quad \min_{x} \sum_{i=1}^{m} (y_i - a_i^T x)^2$$

- Mixed-integer (qp) optimization (quadratic / linear)

$$\min \; c^T x + x^T Q x$$
$$s.t \quad Ax \geq b$$
$$x_j \in \mathbb{Z} \qquad \forall j \in J$$

$$e.g., \quad \min_{x,z} \sum_{i=1}^{m} (y_i - a_i^T x)^2$$

$$s.t \quad -M z_i \leq x_i \leq M z_i \qquad i = 1, \dots, n$$

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2 Solution

Min $f(x)$

s.t. $x \in \mathcal{F}$

if $f(x)$ is a convex function
$\mathcal{F}$ is a convex set.
then local minima = global minima.

e.g. LP (convex)
MLP (non-convex)

3. Unconstrained Minimization

$$\min_{x \in \mathbb{R}^n} f(x)$$

local min $\Rightarrow \nabla f(x) = 0$

if $f(x)$ is convex $\Leftrightarrow$ (differentiable)

Iterate x^k

$$x^{k+1} = x^k + \lambda d^k$$

?

$\nabla f(x^k)^T d^k < 0$

- Steepest descent
  $d^k = -\nabla f(x^k)$
- Newton Methods.
  $d^k = -\nabla^2 f(x^k)^{-1} \nabla f(x^k)$

$$\min_{\lambda \geq 0} f(x^k + \lambda d^k)$$

- Binary Search

$$\min f(x)$$

find $f'(x) = 0$

- Newton Methods.

$$x^{k+1} = x^k - \frac{f'(x^k)}{f''(x^k)}$$

Linear Optimisation

$$\min c^T x = z \quad \text{s.t. } Ax \geq b$$

$\longleftrightarrow$ dual

$$\max y^T b = w \quad \text{s.t. } y^T A \leq c^T, \; y \geq 0$$

Strong dual $z = w$

Optimality Condition

x is optimal for (P) iff.

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Linear optimization

$z = \min c^T x$ (P) s.t. $Ax \geq b$, $x \geq 0$ $\quad \overset{\text{dual}}{\longleftrightarrow} \quad w = \max y^T b$ s.t. $y^T A \leq c^T$, $y \geq 0$

strong duality: $z = w$

optimality conditions

x is optimal for (P) iff:

1. $Ax \geq b$, $x \geq 0$ (PF)

   $\exists\, y$ s.t.

2. $y^T A \leq c^T$, $y \geq 0$ (DF)

3. $\forall j \; x_j (y^T A_j - c_j) = 0 \quad \forall j = 1, \dots, n$

(CS) $y_i (a_i^T x - b_i) = 0 \quad \forall i = 1, \dots, m$

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ISE 530 Optimization Methods for Analytics

Midterm

Instructor: Andrés Gómez

This examination is open book, notes and other handwritten or printed material. No electronic devices allowed, including: cellphones, laptops, calculators. The examination is to be completed individually. It consists of four questions, and the duration is 110 minutes.

Name:
Student ID:

Question 1

The university of Southern California offers the following seven courses in the PhD program in a given year. The name of the courses, credits required and utility (if taken) is described in the following table.

#	**Name**	**Credits**	Utility
1	Linear optimization	4	5
2	Stochastic processes	4	3
3	Large-scale optimization	4	4
4	Quantum computing	4	7
5	Decision analysis	2	2
6	Department seminar	1	2
7	Directed research	1 (can be taken multiple times)	1

Suppose you can take up to 14 units among the offered courses. The problem of choosing the best selection of courses to maximize the utility is given by

$$\begin{aligned}
&\max_{\boldsymbol{x}} 5x_1 + 3x_2 + 4x_3 + 7x_4 + 2x_5 + 2x_6 + x_7 && \text{(Utility)}\\
&\text{s.t. } 4(x_1 + x_2 + x_3 + x_4) + 2x_5 + x_6 + x_7 \leq 14 && \text{(Credit hours)}\\
&\quad x_i \in \{0,1\} && \forall i = 1,\ldots,6 \quad \text{(Binary variables)}\\
&\quad x_7 \in \mathbb{Z}, && \text{(Directed research can be taken multiple times)}
\end{aligned}$$

where x_i indicates whether a course if taken or not (if $1 \leq i \leq 6$) or the number of credits taken (if $i = 7$).

Modify the formulation to account for the following considerations.

1. Quantum computing and decision analysis are scheduled at the same time, so it is not possible to take both courses. $x_4 + x_5 \leq 1$
2. Linear optimization is a prerequisite of large-scale optimization. Thus, to take the later course, linear optimization must also be taken. $x_3 \leq x_1$
3. To ensure breadth of the program, it is not possible to take all courses with "optimization" or "computing" in the name. $x_1 + x_3 + x_4 \leq 2$
4. Students who do not undertake any units of directed research must be enrolled in the department seminar. $x_6 + x_7 \geq 1$
5. Students are required to take at least two of the following three fundamental courses: linear optimization, stochastic processes and decision analysis. $x_1 + x_2 + x_5 \geq 2$

Question 2

1. Samantha has recently started a small business selling handcrafted candles. The fixed costs associated with starting the business, including the purchase of initial equipment and rental of a small workshop space, amount to \$10,000. Each candle she produces costs her \$5 in raw materials and other variable costs. She sells each candle for \$20. She is interested in finding her breakeven point, that is, the number of candles she needs to sell to break even.
   a) Formulate this problem as the problem of finding a zero of a function.
   b) Clearly, if no candles are sold, then the business is losing money. Moreover, if 1,000 candles are sold, then the company is making money. Using this information as starting points, do two iterations of binary search.
2. Consider the same situation above, but now ~~Emily~~ Samantha needs to determine the price at which she sells candles. If she decides to sell candles for p dollars, then the number of units sold is $1,000 - 20p$.
   a) Formulate the problem of maximizing profit (after costs) as an optimization problem.
   b) Starting from $p = 20$, do one iteration of Newton method. Is this solution optimal?
   c) Should Emily continue with her business?

Question 2

1. Samantha has recently started a small business selling handcrafted candles. The fixed costs associated with starting the business, including the purchase of initial equipment and rental of a small workshop space, amount to \$10,000. Each candle she produces costs her \$5 in raw materials and other variable costs. She sells each candle for \$20. She is interested in finding her breakeven point, that is, the number of candles she needs to sell to break even.
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2. Consider the same situation above, but now Emily needs to determine the price at which she sells candles. If she decides to sell candles for p dollars, then the number of units sold is $1,000 - 20p$.
   a) Formulate the problem of maximizing profit (after costs) as an optimization problem.
   b) Starting from $p = 20$, do one iteration of Newton method. Is this solution optimal?
   c) Should Emily continue with her business?

a) $\max_{p \geq 0} \underbrace{-10,000 + (1000 - 20p)(p-5)}_{f(p)}$

(b) $f(p) = -10,000 + 1000p - 5000 + 100p - 20p^2$

$-15,000 + 1100p - 20p^2$

$f'(p) = 1100 - 40p$

$f''(p) = -40$

$p^1 = 20 - \frac{f'(p)}{f''(p)} = 20 - \frac{1100 - 40 \cdot 20}{-40} = 27.6$

$f'(27.5) = 1100 - 40(27.5) = 0$ optimal ✓

$\pi'(p) = 1{,}100 - 40p$

$\pi''(p) = -40$

$p^{k+1} \leftarrow p^k - \frac{\pi'(p)}{\pi''(p)}$

$p^k - \frac{\pi(p)}{\pi'(p)}$

$p^1 \leftarrow 20 - \frac{1{,}100 - 40 \cdot 20}{-40} = 20 \frac{-300}{-40}$

$= 27.5$

$\pi'(27.5) = 1{,}100 - 40(27.5) = 0$ optimal ✓

$40 \times 20 = 800$

$40 \times 7.5 = 4 \times 75 = 300$

c) $\pi(p) = -10{,}000 + (1000 - 20 \times 27.5)(22.5)$

$= -10{,}000 + (1000 - 550)\ 22.5$

$= -10{,}000 + 45 \times 225$

$= -10{,}000 + 225 \times 40 + 225 \times 5$

$= -10{,}000 + 9{,}000\ 1125$

$1125 > 0$

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Question 3

1. Consider the quadratic optimization problem

$$f(\boldsymbol{x}) = 2x_1 + 3x_2 + (x_1 + x_2)^2 + 2x_1^2 + x_2^2.$$

   a) Find a stationary point of this problem. Is it optimal?

   b) Starting at point $\boldsymbol{x} = (11)$, do one iteration of steepest descent.

2. Consider the problem of computing a cubic root of 853.

   a) Do two iterations of binary search. You may assume that there exists a solution x^* such that $1 \le x^* \le 10$.

   b) Do two iterations of Newton method, starting from the point $x^0 = 1$.

a) $\nabla f(x) = \begin{pmatrix} 6x_1 + 2x_2 + 2 \\ 2x_1 + 4x_2 + 3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

$x_1 = \frac{1}{10} \quad x_2 = -\frac{7}{10}$

b is convex, so optimal

b) $\nabla f(x) = \begin{pmatrix} 6x_1 + 2x_2 + 2 \\ 2x_1 + 4x_2 + 3 \end{pmatrix}$

$-\nabla f(1) = \begin{pmatrix} -10 \\ -9 \end{pmatrix}$

$x^1 = x^0 - \lambda \begin{pmatrix} 10 \\ 9 \end{pmatrix}$

(prof 好像 λ or x^0)

$\min_{\lambda} f\left(\begin{pmatrix} 1 \\ 1 \end{pmatrix} - \lambda \begin{pmatrix} 10 \\ 9 \end{pmatrix}\right)$

Question 3

1. Consider the quadratic optimization problem

$$f(\boldsymbol{x}) = 2x_1 + 3x_2 + (x_1 + x_2)^2 + 2x_1^2 + x_2^2.$$

   a) Find a stationary point of this problem. Is it optimal?

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   a) Do two iterations of binary search. You may assume that there exists a solution x^* such that $1 \leq x^* \leq 10$.

   b) Do two iterations of Newton method, starting from the point $x^0 = 1$.

$f(x) = x^3 - 853 = 0.$

a)

a	b	$\frac{a+b}{2} = c$	$f(c)$
1	10	5.5	$(5.5)^3 - 853 < 0$
⋮	⋮	⋮	⋮

(b) $f'(x) = 3x^2$

$$x^{k+1} \leftarrow x^k - \frac{f(x^k)}{f'(x^k)}$$

$$x^1 \leftarrow 1 - \frac{853}{3} = -284$$

$$x^2 \leftarrow -285 - \ldots$$

Question 4

1. Consider the optimization problem

$$\begin{aligned} \min\ & x_1 + 3x_2 + 4x_3 + x_4 \\ \text{s.t. } & 2x_1 + 2x_2 + x_4 = 2 \\ & x_1 + x_2 + x_3 + x_4 \geq 5 \\ & x_1, x_2, x_3, x_4 \geq 0 \end{aligned}$$

   Compute its dual.

2. Consider the optimization problem

$$\begin{aligned} \max\ & 2x_1 + x_2 - x_3 \\ \text{s.t. } & x_1 + 2x_2 + x_3 \leq 8 \\ & -x_1 + x_2 - 2x_3 \leq 4 \\ & x_1, x_2, x_3 \geq 0, \end{aligned}$$

   and its dual

$$\begin{aligned} \min\ & 8y_1 + 4y_2 \\ \text{s.t. } & y_1 - y_2 \geq 2 \\ & 2y_1 + y_2 \geq 1 \\ & y_1 - 2y_2 \geq -1 \\ & y_1, y_2 \geq 0. \end{aligned}$$

1. Write the optimality conditions.
2. The optimal dual solution is $y_1 = 2$, $y_2 = 0$. Compute an associated primal solution using the optimality conditions.

(P) max $2y_1 + 5y_2$

s.t. $2y_1 + y_2 \leq 1$

$y_1 + y_2 \leq 3$

$y_2 \leq 4$

$y_1 + y_2 \leq 1$

y_1 free

$y_2 \geq 0$

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PF

$$-x_1 + x_2 - 2x_3$$

$$x_1, x_2, x_3 \geq 0$$

$$2y_1 + y_2 \geq 1$$

$$y_1 - 2y_2 \geq -1$$

$$y_1, y_2 \geq 0$$

CS +1

$$x_1(y_1 - y_2 - 2) = 0$$

$$x_2(2y_1 + y_2 - 1) = 0$$

$$x_3(y_1 - 2y_2 + 1) = 0$$

$$y_1(x_1 + 2x_2 + x_3 - 8) = 0$$

$$y_2(-x_1 + x_2 - 2x_3 - 4) = 0$$

Mistake

complementary conditions

if $x_i > 0$ optimal then (PF)

1. $Ax \geq b$ $x \geq 0$ (PF)
2. $y \geq 0$ $yA \leq c$, $y \geq 0$ (DF)
3. $\forall i$ $x_i(yA_i - y) = 0$ $y_i(x_i - b_i) = 0$ $\forall i = 1, \ldots, m$

(cs) $y_i(a_i x - b_i) = 0$ $\forall i = 1, \ldots, m$

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a) write optimality +1

PF
$$x_1 + 2x_2 + x_3 \le 8$$
$$-x_1 + x_2 - 2x_3 \le 4$$
$$x_1, x_2, x_3 \ge 0$$

DF
$$y_1 - y_2 \ge 2$$
$$2y_1 + y_2 \ge 1$$
$$y_1 - 2y_2 \ge -1$$
$$y_1, y_2 \ge 0$$

CS +1

$$x_1 (y_1 - y_2 - 2) = 0 \quad \checkmark$$
$$x_2 (2y_1 + y_2 - 1) = 0 \quad (x_2 = 0)$$
$$x_3 (y_1 - 2y_2 + 1) = 0 \quad (x_3 = 0)$$
$$y_1 (x_1 + 2x_2 + x_3 - 8) = 0$$
$$y_2 (\quad -4) = 0 \quad \checkmark$$

$x_1 + 2x_2 + x_3 = 8$

$(x_1 = 8)$

Mostafa

optimality conditions

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